

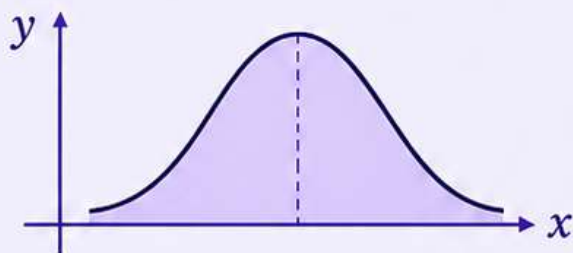
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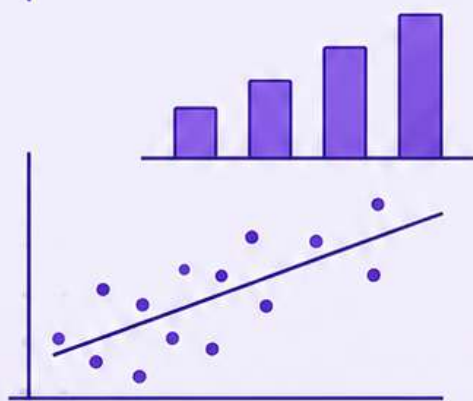
Mathematics & Statistics for Data Analyst

Build Strong Foundations.
Solve Real-World Problems.

$$\sum_{i=1}^n x_i$$



$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}$$



$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$



Mathematical
Concepts



Statistical
Methods



Data Analysis
Techniques



Problem Solving
Skills



Master the Basics. Excel in Analytics.

Table of Contents

12 Units • 100 Topics • A Complete Statistics Journey

Unit 1 Descriptive Statistics (Summarizing & Describing Data) 16 – 37

1.1 Measures of center

- i) Mean, Median & Mode
- ii) Trimmed & Winsorized Mean
- iii) Midrange & Mid-quartile

1.2 Measures of spread

- i) Range & Interquartile Range
- ii) Variance & Standard Deviation
- iii) Mean Absolute Deviation (MAD)
- iv) Coefficient of Variation (CV)

1.3 Shape & position

- i) Skewness (Pearson & Fisher)
- ii) Kurtosis & Excess Kurtosis
- iii) Percentiles, Quartiles & Deciles
- iv) Z-scores & Standardisation

1.4 Visualisation

- i) Histograms & Bin Width Selection
- ii) Box plots & Violin plots
- iii) Stem-and-leaf & QQ plots
- iv) Frequency & Contingency Tables

Unit 2 Probability Foundations 38 – 66

2.1 Core concepts

- i) Sample Spaces & Events
- ii) Classical, Empirical & Subjective Probability
- iii) Kolmogorov Axioms
- iv) Complement & Inclusion-Exclusion

2.2 Rules & operations

- i) Addition Rule

- ii) Multiplication Rule
- iii) Conditional Probability $P(A|B)$
- iv) Law of Total Probability

2.3 Counting & combinatorics

- i) Fundamental Counting Principle
- ii) Permutations (ordered)
- iii) Combinations (unordered)
- iv) Probability Trees & Venn Diagrams

2.4 Random variables

- i) Discrete vs Continuous RVs
- ii) Expected Value $E(X)$
- iii) Variance $\text{Var}(X)$ & Covariance
- iv) Moment Generating Functions

Unit 3 Probability Distributions67 – 101

3.1 Discrete distributions

- i) Bernoulli Distribution
- ii) Binomial Distribution
- iii) Poisson Distribution
- iv) Geometric Distribution
- v) Negative Binomial
- vi) Hypergeometric Distribution

3.2 Continuous distributions

- i) Uniform Distribution
- ii) Exponential Distribution
- iii) Gamma Distribution
- iv) Beta Distribution
- v) Log-Normal Distribution

3.3 Sampling distributions

- i) Student's t-Distribution
- ii) Chi-Square Distribution
- iii) F-Distribution
- iv) Distribution Relationships

3.4 Selection & application

- i) Distribution Selection Framework
- ii) Fitting Distributions to Data
- iii) Mixture Distributions

Unit 4 Normal Distribution 102 – 137

4.1 Core properties

- i) Definition & Parameters (μ , σ)
- ii) Standard Normal Distribution $Z \sim N(0,1)$
- iii) Empirical Rule (68-95-99.7) iv) Normal PDF & CDF

4.2 Z-scores & computation

- i) Z-score Interpretation
- ii) Finding Areas Under the Curve
- iii) Inverse Normal (Quantile Function)

4.3 Checking normality

- i) QQ Plots (Quantile-Quantile)
- ii) Shapiro-Wilk Test
- iii) D'Agostino-Pearson Test iv) Anderson-Darling Test

4.4 Extensions

- i) Bivariate Normal Distribution
- ii) Multivariate Normal Distribution
- iii) Log-Normal Connection
- iv) Transformations to Achieve Normality

Unit 5 Central Limit Theorem 138 – 171

5.1 Foundations

- i) Law of Large Numbers (Weak & Strong)
- ii) Sampling Distributions
- iii) Standard Error (SE)
- iv) Bias-Variance of Estimators

5.2 CLT statement & proof

- i) CLT Formal Statement
- ii) Proof Intuition via MGFs

iii) Berry-Esseen Theorem

iv) CLT for Proportions

5.3 Applications

i) CLT in Practice — Required Sample Size

ii) CLT with Non-Normal Populations

iii) Finite Population Correction

iv) Delta Method

Unit 6 Hypothesis Testing 172 – 211

6.1 Framework

i) Null & Alternative Hypotheses (H_0 , H_1)

ii) Type I Error (α) — False Positive

iii) Type II Error (β) — False Negative

iv) Power ($1-\beta$)

v) p-value — Definition & Misinterpretation

6.2 Parametric tests

i) One-Sample z-test & t-test

ii) Independent Samples t-test

iii) Paired Samples t-test

iv) Chi-Square Goodness of Fit

v) Chi-Square Test of Independence vi) F-test for Equal Variances

6.3 Effect size & power

i) Cohen's d (mean differences)

ii) Eta-squared & Omega-squared

iii) Sample Size Calculation

iv) Post-hoc Power Analysis

6.4 Multiple testing

i) Family-Wise Error Rate (FWER)

ii) Bonferroni Correction

Unit 7 Confidence Intervals 212 – 247

7.1 Core concept

i) Point vs Interval Estimation

- ii) Confidence Level vs Significance Level
- iii) Correct Interpretation of CIs
- iv) Margin of Error

7.2 CIs for specific parameters

- i) CI for Population Mean (z-interval)
- ii) CI for Population Mean (t-interval)
- iii) CI for Proportion
- iv) CI for Variance (Chi-Square)
- v) CI for Difference of Two Means
- vi) CI for Odds Ratio & Relative Risk

7.3 Advanced topics

- i) CI Width & Sample Size
- ii) Bootstrap Confidence Intervals
- iii) Bayesian Credible Intervals
- iv) Simultaneous CIs — Bonferroni & Scheffé

Unit 8 A/B Testing 248 – 295

8.1 Design principles

- i) Randomised Controlled Experiments
- ii) Control vs Treatment Groups
- iii) Randomisation Unit
- iv) A/A Testing

8.2 Sample size & power

- i) Minimum Detectable Effect (MDE)
- ii) Sample Size Calculation
- iii) Stopping Rules & Experiment Duration
- iv) Variance Reduction Techniques

8.3 Analysis & pitfalls

- i) Statistical Significance in A/B Tests
- ii) The Peeking Problem
- iii) Sequential Testing & Always-Valid p-values
- iv) Network Effects & Interference
- v) Novelty & Primacy Effects

8.4 Advanced methods

- i) Multi-Armed Bandit Algorithms
- ii) Bayesian A/B Testing
- iii) Multivariate Testing (MVT)
- iv) Holdout Groups & Long-Run Effects

Unit 9 Correlation & Regression 296 – 339

9.1 Correlation

- i) Pearson Correlation (r)
- ii) Spearman Rank Correlation
- iii) Kendall's Tau
- iv) Partial Correlation
- v) Correlation vs Causation

9.2 Simple linear regression

- i) OLS — Ordinary Least Squares
- ii) R-squared & Adjusted R-squared
- iii) BLUE Theorem (Gauss-Markov)
- iv) Residual Analysis
- v) Influential Points — Leverage & Cook's D

9.3 Multiple regression

- i) Multiple Linear Regression
- ii) Multicollinearity & VIF
- iii) Polynomial & Interaction Terms
- iv) Variable Selection
- v) Regularisation — Ridge & Lasso

9.4 Beyond OLS

- i) Logistic Regression
- ii) Multinomial & Ordinal Logistic Regression
- iii) Poisson Regression
- iv) Model Evaluation Metrics

Unit 10 Bayes Theorem Basics 340 – 376

10.1 Core theorem

- i) Bayes' Formula Derivation
- ii) Prior, Likelihood & Posterior

iii) Bayesian Updating

iv) Bayes vs Frequentist Paradigm

10.2 Conjugate priors & computation

i) Beta-Binomial Conjugacy

ii) Gamma-Poisson Conjugacy

iii) Normal-Normal Conjugacy

iv) Non-Conjugate Priors & MCMC

10.3 Applications

i) Naive Bayes Classifier

ii) Bayesian Hypothesis Testing

iii) Credible Intervals vs Confidence Intervals

iv) Bayesian Networks

Unit 11 Analysis of Variance (ANOVA)377 – 416

11.1 One-Way ANOVA

i) Between-Group vs Within-Group Variance

ii) F-Statistic & F-Distribution

iii) ANOVA Assumptions iv) Effect Size — Eta-squared & Omega-squared

11.2 Post-hoc tests

i) Tukey's HSD

ii) Bonferroni Correction in ANOVA

iii) Scheffé's Method iv) Dunnett's Test

11.3 Extensions

i) Two-Way ANOVA

ii) Interaction Effects

iii) Repeated Measures ANOVA iv) MANOVA Basics v) Welch ANOVA

Unit 12 Time Series Statistics417 – 463

12.1 Decomposition & structure

i) Time Series Components

ii) Additive vs Multiplicative Decomposition

iii) Autocorrelation Function (ACF) iv) Partial ACF (PACF)

12.2 Stationarity

- i) Stationarity — Definition
- ii) Augmented Dickey-Fuller (ADF) Test
- iii) KPSS Test iv) Making Series Stationary

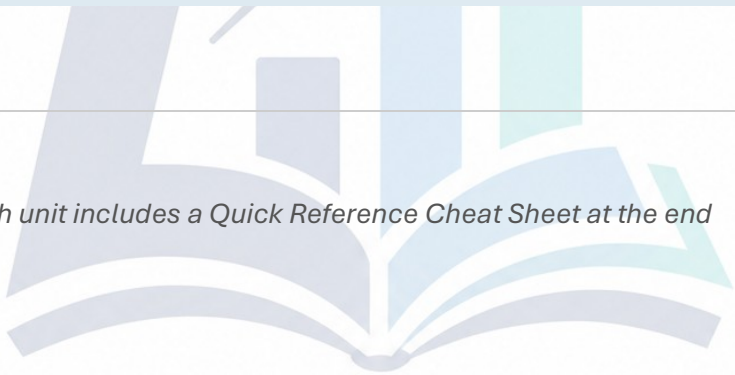
12.3 Classical models

- i) Moving Averages & Exponential Smoothing
- ii) AR, MA, ARMA Models
- iii) ARIMA & SARIMA iv) ARCH & GARCH (Volatility Modelling)

12.4 Evaluation & modern methods

- i) Train-Test Split for Time Series
- ii) Forecast Evaluation Metrics
- iii) Prophet & ML-Based Forecasting iv) Granger Causality

Each unit includes a Quick Reference Cheat Sheet at the end



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.handbook

LEARN SMART. ANALYZE BETTER. GROW FASTER.